



OMERTA AGENT

Features & Full Usage Guide

An offline-capable coding & firmware agent that asks before it acts.

Runs on Android (no Termux), Linux, Kali, macOS, Windows, and any browser on your LAN.

Swappable brains: local models offline, frontier models online.

Version 1.1.0

always-ask · offline+online · unlimited chats · self-contained APK

What OMERTA is

A complete agent framework that drives real backends. It is not a clone of Codex or Claude — model weights can't be copied out of the labs that made them — but it **calls** them (and any local model), with one hard rule: it proposes every action and runs nothing until you approve it. That rule is enforced in control flow, not in a prompt, and is covered by tests.

The Android app is fully self-contained: it embeds Python and runs the agent in-process, so there is no Termux and nothing external to set up. The backend is a normal HTTP service — no websocket, no dependency to install — and the same code runs on every platform.

Feature matrix

Capability	What you get
Offline + online	Local GGUF models (Ollama / llama.cpp / LM Studio) and 6 cloud providers, auto-switching, with an explicit Offline / Online / Auto mode.
Unlimited chats	No message cap. Long conversations are compacted into a bounded context window so they never hit a wall. Truly unlimited offline.
Always ask first	Every command, file write, plugin side effect and connector call is proposed and waits for your yes. Enforced in code.
Learns your choices	Every approve / deny / edit is recorded and fed back; denials become standing preferences.
Sandbox	Tiered execution, a hard-deny list that can't be approved, and automatic backups before edits.
Network safety	Per-install token auth, loopback-exempt, spoof-proof against forwarded-header tricks.
Self-contained mobile	Android APK embeds CPython (Chaquopy); the agent runs on-device with no Termux.
Connectors / plugins / skills	MCP client, Python plugin loader, keyword-loaded skill playbooks.
Shared memory	Peer (LAN) and shared-folder sync; denials and deletions propagate across devices.

Architecture

One agent, many front-ends

All request logic lives in `core/api.py`, so there is exactly one approval gate and one wire protocol no matter how you reach the agent:

- **CLI** — `omerta` in a terminal.
- **FastAPI server** (`server.py`) — for desktop / LAN, reachable from any browser.
- **Stdlib server** (`core/httpd.py`) — zero third-party deps; what the Android app runs, and a drop-in fallback anywhere FastAPI isn't installed.
- **Electron** desktop shell and the responsive **web UI**.

Embedded Android backend

Chaquopy bundles CPython 3.11 plus pure-Python deps (`requests`, `pyyaml`) into the APK. A foreground service starts the stdlib server on `127.0.0.1`; the WebView loads the same UI. The agent's Python is the repo's `core/` and `tools/`, staged into the APK at build time and extracted on first launch — so the approval gate and deny-list can't drift from desktop.

```
MainActivity → BackendService → Chaquopy Python → omerta_boot → omerta_android → core.httpd
(127.0.0.1) ■■■ WebView (loopback, no token)
```

Install & run

Android (no Termux)

Install the APK, open it, and pick a brain in ■ → **MODE / MODEL ACCESS** — paste a cloud API key (online) or point it at a local model (offline). Nothing else to install.

pip — Linux / macOS / Windows / Termux

```
pip install omerta_agent-1.0.0-py3-none-any.whl[all]
omerta # interactive CLI
omerta serve # web UI on :8787
omerta doctor # what works, what's missing
omerta sync # share memory across devices
```

Give it a brain

```
export ANTHROPIC_API_KEY=sk-ant-... # online
ollama serve && ollama pull qwen2.5-coder:7b # offline, unlimited
```

The always-ask guarantee

Every side-effecting action is proposed, never executed, until you answer. What you approve is character-for-character what runs.

Answer	What happens
RUN	Executes, is logged, and is remembered as approved.
EDIT	Your rewrite runs instead — a strong signal; it learns your form.
REFUSE	Never retried; the agent proposes a different approach.

Risk tiers & the hard-deny list

- **DENY** — patterns like `rm -rf /`, `fastboot flashall -w`, `mkfs` on a raw disk: refused outright, not runnable even if you try to approve.
- **HIGH_RISK** — flashing, `dd`, `git push --force`, `rm -rf`: shown with a loud destructive warning.
- **NORMAL / LOW_RISK** — everything else; read-only commands run freely so the agent can actually think.

It generalizes: denying `fastboot erase userdata` teaches it about `fastboot erase` broadly. `/prefs` shows what it thinks it knows.

Offline / Online / Auto & unlimited chats

Mode	Behavior
Auto	Online if reachable, otherwise a local model — switches on its own if you lose signal mid-task.
Offline	Local models only. Never touches the network. True unlimited — your hardware, your rules.
Online	Cloud providers only.

Chats have no message cap. To keep an endless conversation from overflowing the model's context window, only the most recent turns are replayed; older turns are already distilled into long-term memory, so nothing durable is lost.

Model providers

Provider	Kind	Net	Default model
claude / claude-opus	Anthropic	online	claude-sonnet-4-6 / opus-4-1
openai	OpenAI	online	gpt-4o
openrouter	OpenAI-compat	online	anthropic/claude-sonnet-4
groq	OpenAI-compat	online	llama-3.3-70b-versatile
gemini	OpenAI-compat	online	gemini-2.0-flash
ollama	Ollama	offline	qwen2.5-coder:7b
llamacpp	llama.cpp	offline	local GGUF

Provider	Kind	Net	Default model
lmstudio	OpenAI-compatible	offline	local model

The Anthropic provider works without the SDK (pure requests), so Termux needs no Rust toolchain for pydantic-core.

Extending it

Skills

Playbooks in `skills/<name>/SKILL.md`, auto-loaded by keyword. Ships with eight: `android-build`, `firmware-flash`, `cross-compile`, `apk-analysis`, `termux-env`, `git-hygiene`, `debug-build-failure`, `release-packaging`.

Plugins

Drop a Python file in `plugins/` exposing a `register()` that returns tools; ~20 lines. Any tool marked `side_effects=True` is routed through the same approval gate — plugins cannot bypass `always-ask`.

Connectors (MCP)

Any MCP server, stdio or HTTP, configured in `connectors.yaml` (GitHub, filesystem, git, sqlite, Sentry, Cloudflare presets). Tools appear as `mcp.<server>.<tool>` and every call needs approval.

Importing what you already have

```
/import ~/Downloads/conversations.json # claude.ai or ChatGPT export
/import ~/notes # a markdown directory
/import ~/src/omerta-beats # a repo — learns its structure
```

One brain across every device

What your phone learns, your Kali box knows — with no cloud account in the middle. Denials travel too: refusing a command on one device stops it being proposed on the others.

```
export OMERTA_SYNC_DIR=~/.Syncthing/omerta # shared folder (works offline)
/sync # sync via the shared folder
/sync 192.168.1.42:8787 # straight to another device on the LAN
/sync status # peers, folders, last sync
```

Rows carry a uid so re-syncing is idempotent; facts merge last-write-wins; tombstoned deletions propagate and are never resurrected; a DENIED choice is never dropped in a conflict — safety signals only accumulate.

Security & audit

- **Model can't act alone** — every side effect suspends for approval; verified by tests.
- **Hard-deny list** — destructive patterns can't be approved at all.
- **Token auth** on the network server; loopback exempt unless the request carries forwarding headers (spoof-proof).
uvicorn runs with proxy headers off.
- **On-device secrets** — API keys stored 0600 in app-private storage, never leave the device; the write path is loopback-only and off by default on LAN servers.
- **Auto-backups** before any file edit; checkpoints are restorable.
- **Per-project locking** in the threaded server; 16 MiB request cap.

Full threat model, findings and accepted design choices: docs/AUDIT.md. Re-run the checks with `bash tests/run_all.sh` (approval, deny-list, auth spoofing on both servers, tool-parse, mode routing, unlimited-chat trim, multi-device sync).

Beyond chat — the engineering toolkit

OMERTA is a workflow system, not just a chat box. These verbs work from the CLI and, where they change the world, still pass through the approval gate.

Codebase intelligence

`omerta index` builds a real index — Python parsed with `ast` (classes, functions, methods + imports + a call graph), other languages via conservative regex. Then `search / symbol / deps / calls` answer “where is this defined / imported / called” instantly. The same power is available to the agent as the **code_index** plugin tools.

Engineering roles

Ten focus profiles steer the one agent under the same gate: `omerta plan` (Architect), `review` (Reviewer, read-only), `build / test / debug`, and `Firmware / Security / Docs / Researcher / Developer`. `omerta role <name>` sets a default. `omerta workflow` chains roles (`plan→review`) or runs a **team concurrently** (`workflow team`) — thread-isolated, read-only, aggregating independent findings.

Sandbox & recovery

`omerta sandbox snapshot` copies a workspace before a risky change and `rollback` restores it (edited and deleted files). Opt-in isolation (`OMERTA_ISOLATE=1`) wraps commands with resource limits and, when `bubblewrap/firejail` is installed, denies the network and shadows key stores (`~/.ssh`, `~/.aws`, `~/.gnupg`).

Learning

`omerta teach "rule"` stores a durable project rule; `memory show / rules / forget` manage it. Approvals/denials are already learned automatically.

Packaging

Ship it as a wheel, an **OCI/Docker image** (`docker/Dockerfile`), a validated **.deb** (`packaging/build-deb.sh`), a thin `AppImage`, the self-contained `APK`, or `Electron` installers. `scripts/build_all.sh` builds whatever the host can.

HTTP API reference

Method & path	Purpose
POST <code>/api/chat</code>	One chat turn: {project, kind, text cmd note} → full result.
GET <code>/api/status</code>	Providers, active model, mode, memory, skills, plugins, connectors.
POST <code>/api/model</code>	Set the active provider (or 'auto').
POST <code>/api/mode</code>	Set network mode: auto offline online.
GET/POST <code>/api/secret</code>	Read which keys are set (booleans only) / set one (loopback-only).
GET <code>/api/memory</code>	Search long-term memory + learned preferences.
GET <code>/api/history</code>	Recent command log.

Method & path	Purpose
GET /api/sync/pull, POST /api/sync/push	Peer memory exchange.
POST /api/sync/run	Trigger a sync ({peer} or {dir}).
POST /api/plugins/reload, /api/connectors/reconnect	Hot-reload.

CLI / one-shot

```

omerta # interactive
omerta -c "audit this repo" # one-shot
omerta --project omerta-beats
omerta serve | doctor | sync | version
omerta teach "rule" | memory show | rules | role <name>
omerta index | search "q" | symbol Name | deps | calls Name
omerta git status|diff|log|review | evidence add <conf> "claim"
omerta plan/review/build/test/debug "... " | sandbox snapshot|rollback
omerta workflow <pipeline> "task"
omerta firmware inspect <img> | analyze <dtb|dtbo|boot.img> | extract <img> | report <dir>

```

Firmware & device bring-up

OMERTA treats firmware the way a careful engineer does: evidence first, no invented hardware values. Every field is tagged **CONFIRMED** > **LIKELY** > **INFERRED** > **UNKNOWN**; UNKNOWN is a valid answer, so it says “UNKNOWN — evidence required” rather than guess a GPIO, regulator or panel timing.

Read-only tool (Level 0)	What it does
inspect_image(path)	Identify a boot / vendor_boot / dtbo image or raw .dtb; report header version (v0–v4), sizes, cmdline, os_version, and dt-table entries.
analyze_dtb(path)	Decode a device tree with a pure-Python FDT parser (no dtc) — a raw .dtb, a dtbo/dt_table entry, or the DTB embedded in a boot.img — into model/SoC/CPU/display/touch/regulators, each with a confidence.
board_report(dir)	Fuse an adb evidence dir (getprop/cpuinfo/meminfo/partitions/dmesg + optional .dtb) into a BOARD REPORT with per-field confidence.
collect_evidence_plan()	The ordered, read-only adb commands to gather evidence.

Safety levels map onto the approval gate: L0 read runs freely; L1 build / L2 modify / L3 device I/O are proposed and approved; L4 destructive (flash/erase/format) needs explicit approval and the hard-deny list blocks the unrecoverable ones outright. `firmware/t509k/` is a discovery-only TCL T509K (MT6765) starter tree, and the **firmware-bringup** skill drives the STOCK → EXTRACT → ANALYZE → RECONSTRUCT → BUILD → BOOT-TEST loop, one subsystem at a time.

Build & release

```

bash scripts/build_all.sh # wheel, standalone binary, Electron, APK
cd android-native && gradle assembleDebug # self-contained APK

```

Cross-platform limits are real: PyInstaller can't cross-compile, macOS signing is mac-only, and the APK needs the Android SDK + a host Python 3.8–3.13 (Chaquopy runs pip on the host). See `INSTALL.md` and `android-native/README.md`.

OMERTA AGENT v1.1.0 — offline-capable coding & firmware agent that asks before it acts. This guide is generated from the repository docs.